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COMPUTER ARCHTECTURE {ADVANCED)WK11

1. Research and Compare: NVIDIA Hopper GPU Architecture

Introduction

The NVIDIA Hopper architecture is a modern Graphics Processing Unit (GPU) architecture developed by NVIDIA and introduced in 2022. It is designed mainly for artificial intelligence (AI), machine learning, deep learning, and high-performance computing (HPC). One of the GPUs based on this architecture is the NVIDIA H100 Tensor Core GPU, which provides high processing power for complex calculations and large datasets. [1]

Key Architectural Features

The NVIDIA H100 uses the Hopper architecture and contains up to 132 Streaming Multiprocessors (SMs) in its SXM version. Each SM includes 128 FP32 CUDA cores and four fourth-generation Tensor Cores. These processing units allow the GPU to execute many operations in parallel. Tensor Cores are particularly useful for accelerating matrix calculations, which are widely used in AI training and inference. The H100 SXM version therefore contains 16,896 FP32 CUDA cores and 528 Tensor Cores. [1]

Another important feature is the Transformer Engine, which supports FP8 and FP16 numerical formats to accelerate transformer-based AI models. This helps improve the speed of applications such as language models, image recognition, and other deep-learning systems. Hopper also introduces the Tensor Memory Accelerator (TMA), which improves the movement of large data blocks between global memory and shared memory.

Memory Hierarchy

The H100 has a memory hierarchy consisting of registers, shared memory, L1 cache, L2 cache, and high-bandwidth GPU memory. Registers provide very fast storage for individual threads, while shared memory allows threads within an SM to exchange data efficiently. The L1 cache helps reduce the time required to access frequently used data.

The GPU also includes a 50 MB L2 cache, which is shared across the GPU's processing units. Its high-bandwidth memory (HBM3) provides up to 80 GB of capacity and approximately 3 TB/s of memory bandwidth in the SXM version. This allows the GPU to process large datasets quickly. The Tensor Memory Accelerator further improves efficiency by transferring multidimensional data with less overhead. [2]

Target Application Domain

The NVIDIA Hopper architecture primarily targets data-centre AI and high-performance computing rather than gaming. It is used to train and run large language models, perform scientific simulations, analyse big datasets, and support medical and engineering research. Its high-speed NVLink interconnect also allows multiple GPUs to communicate efficiently in large computing systems.

Conclusion

In conclusion, NVIDIA Hopper represents an important development in GPU technology. Its 132 SMs in the H100 SXM version, advanced Tensor Cores, large memory capacity, and high memory bandwidth make it suitable for demanding AI and HPC workloads. Compared with gaming-focused GPU architectures, Hopper prioritises parallel computing, data processing, and scientific calculations. These features make it valuable for modern data centres and research institutions.

References

1. NVIDIA. (2022). *NVIDIA Hopper Architecture*. https://www.nvidia.com/en-us/data-center/technologies/hopper-architecture/
2. NVIDIA. *NVIDIA H100 Tensor Core GPU Architecture Whitepaper*. https://resources.nvidia.com/en-us-hopper-architecture

2. Research and Compare: NVIDIA Hopper GPU Architecture

Introduction

The NVIDIA Hopper architecture is a modern Graphics Processing Unit (GPU) architecture introduced by NVIDIA in 2022. It is designed mainly for artificial intelligence (AI), machine learning, deep learning, and high-performance computing (HPC). The NVIDIA H100 Tensor Core GPU is one of the main products based on this architecture. It provides high processing power for complex calculations and large datasets. [1]

Key Architectural Features

The NVIDIA H100 SXM version contains 132 Streaming Multiprocessors (SMs). Each SM has 128 FP32 CUDA cores and four fourth-generation Tensor Cores. This gives the GPU 16,896 FP32 CUDA cores and 528 Tensor Cores. These processing units allow thousands of calculations to be performed in parallel, improving performance in computationally demanding applications. [1]

An important feature is the Transformer Engine, which accelerates AI models using FP8 and FP16 numerical formats. This improves the speed and efficiency of training and running transformer-based models, including large language models. Another feature is the Tensor Memory Accelerator (TMA), which transfers large blocks of data between GPU memory and shared memory while reducing the workload on processing threads. [2]

Memory Hierarchy

The Hopper architecture uses a memory hierarchy to improve data access and processing speed. At the top are registers, which provide very fast storage for individual threads. Shared memory allows threads within an SM to exchange data quickly, while the L1 cache stores frequently accessed data. The L2 cache is shared across the GPU and reduces the need to repeatedly access external memory.

The H100 SXM version has a 50 MB L2 cache and 80 GB of HBM3 high-bandwidth memory. Its memory bandwidth reaches approximately 3 TB/s. This enables the GPU to transfer large amounts of data quickly, which is important when processing complex AI models and scientific datasets. [1][2]

Target Application Domain

The main application areas of NVIDIA Hopper are AI, HPC, and data analytics. In AI, it supports training and running large language models, image recognition systems, and deep-learning applications. In HPC, it helps researchers perform scientific simulations, weather modelling, engineering calculations, and genomic analysis. Hopper also supports large-scale computing systems through NVLink, which allows multiple GPUs to communicate at high speeds.

Unlike gaming-focused GPUs, Hopper prioritises data-centre computing and scientific workloads rather than graphics rendering. [1]

Conclusion

In conclusion, NVIDIA Hopper is an advanced GPU architecture designed for demanding computational tasks. Its 132 SMs, fourth-generation Tensor Cores, efficient memory hierarchy, and high-bandwidth HBM3 memory make it suitable for AI and HPC applications. These features allow it to process large datasets efficiently and accelerate calculations that would take much longer on traditional processors.

References

[1] NVIDIA. *NVIDIA Hopper Architecture In-Depth*. https://developer.nvidia.com/blog/nvidia-hopper-architecture-in-depth/

[2] NVIDIA. *NVIDIA H100 Tensor Core GPU Architecture Whitepaper*. https://resources.nvidia.com/en-us-hopper-architecture